A Novel Method to Manage Production on Industry 4.0: Forecasting Overall Equipment Efficiency by Time Series with Topological Features

Korkut Anapa[1†], Ismail Guzel[2*], Ceylan Yozgatligil[3†]

[1*]Institute of Applied Mathematics, Middle East Technical University, Universiteler Mahallesi, Dumlupinar Bulvari No:1, Ankara, 06800, , Turkey.

[2]Network Technologies Department, TUBITAK ULAKBIM, Universiteler Mahallesi, Dumlupinar Bulvari No:1, Ankara, 06800, , Turkey.

[3]Department of Statistics, Middle East Technical University, Universiteler Mahallesi, Dumlupinar Bulvari No:1, Ankara, 06800, , Turkey.

*Corresponding author(s). E-mail(s): iguzel@metu.edu.tr;
Contributing authors: korkut.anapa@metu.edu.tr; ceylan@metu.edu.tr;
†These authors contributed equally to this work.

Abstract

Purpose: Overall equipment efficiency is a key performance indicator in modern manufacturing, yet its volatile nature poses significant challenges for accurate short-term forecasting. This study introduces a novel predictive framework that integrates time series decomposition with topological data analysis to enhance overall equipment efficiency forecasting for different production equipment like hydraulic press systems

Methods: Our methodology begins by decomposing the hourly overall equipment efficiency data into trend, seasonal, and residual components, isolating the complex short-term fluctuations inherent in production environments. The residual component is modelled using a seasonal autoregressive integrated moving average with exogenous approach, which is further enhanced by incorporating exogenous variables derived from statistical measures and topological features extracted from related time series. To effectively manage the high-dimensional

feature space, we propose an integrated feature selection strategy that leverages recursive feature elimination based on seasonal autoregressive integrated moving average with exogenous varioables which are statistically significant, in conjunction with BIC-guided particle swarm optimization. The proposed model was evaluated on datasets collected from different manufacturing systems, where it consistently outperformed conventional forecasting models and advanced transformer-based methods, as evidenced by significant reductions in mean absolute error and mean absolute percentage error.
Results: The results demonstrate that the fusion of classical time series forecasting with topological data analysis improves prediction accuracy and supports proactive maintenance and optimized production teams.
Conclusion: This integrated approach provides a robust framework for addressing the challenges posed by volatile overall equipment efficiency data, paving the way for enhanced operational efficiency and data-driven decision-making in industrial environments.

Keywords: Time Series Forecasting, Time Series Decomposition, Topological Data Analysis, Feature Selection, SARIMA

1 Introduction

1.1 Motivation

Overall Equipment Efficiency (OEE) has become a key metric for assessing and enhancing operational efficiency in modern manufacturing systems. As explained by Nakajima [1], OEE integrates three fundamental components: availability, performance, and quality. This comprehensive measure not only helps identify and categorize productivity losses but also facilitates process improvements by providing actionable insights into resource utilization and equipment performance. Numerous studies have demonstrated the role of OEE as a foundational tool for monitoring and optimizing manufacturing processes, including machines, manufacturing cells, and assembly lines [2, 3]. Through systematic analysis of operational data and the identification of key inefficiencies, OEE supports continuous improvement efforts, aligning manufacturing systems with world-class standards. Moreover, its simplicity and accessibility empower both operators and managers to collaboratively optimize maintenance strategies, minimize breakdowns, and enhance overall productivity [4].

The emergence of Industry 4.0 has further revolutionized OEE measurement and analysis through the integration of interconnected devices, sensors, and advanced data analytics. Today, optimizing maintenance strategies is essential for ensuring operational efficiency. While a limited number of studies have attempted to predict OEE values [5–7], the development of comprehensive production management strategies remains challenging. This is primarily due to the accuracy limitations in OEE prediction, which hinder the establishment of precise, data-driven decision-making frameworks for production management.

By leveraging historical data and real-time sensor information, manufacturers can forecast OEE trends, enabling predictive maintenance strategies that proactively mitigate potential failures [8]. Time-series analysis plays a pivotal role in this approach, allowing the examination of data collected at specific intervals to identify trends, seasonal fluctuations, and cyclical behaviors in equipment performance. Techniques such as AutoRegressive Integrated Moving Average (ARIMA), exponential smoothing, and seasonal decomposition enable the modeling of time-dependent patterns, facilitating precise maintenance scheduling and anomaly detection [9, 10]. These predictive capabilities help reduce unplanned downtime, lower maintenance costs, and extend equipment lifespan, ultimately improving productivity and operational efficiency. Additionally, the integration of Artificial Intelligence (AI) and Machine Learning (ML) enhances predictive accuracy, optimizing production processes and ensuring a competitive edge in dynamic markets [8].

In modern manufacturing environments, OEE data often exhibits high variability due to rapid advancements and continuous improvements in production processes. This underscores the need for short-term data analysis to accurately capture and respond to swift changes in equipment performance. Implementing real-time OEE monitoring systems enables manufacturers to quickly detect and address inefficiencies, leading to improved operational performance [11]. Furthermore, machine learning and data analytics facilitate the development of predictive models that anticipate fluctuations in equipment efficiency, allowing for proactive maintenance and optimization strategies [12]. By focusing on short-term data analysis, organizations can effectively navigate the inherent volatility in OEE metrics, ensuring sustained productivity and competitiveness despite the rapid evolution of production technologies [11, 12].

Forecasting time-series data that is highly volatile and short-term presents significant challenges due to the presence of both chaotic and non-chaotic components. Reliable and consistent predictions often require extensive datasets, yet traditional methods may struggle to extract meaningful patterns from noisy and dynamic time series. Our motivation for this study is to develop a novel method that addresses these challenges and offers the following key benefits:

- **Improved Predictive Maintenance:** By forecasting dips or fluctuations in OEE in advance, maintenance teams can take proactive measures. Since OEE declines often signal impending machine failures or inefficiencies, accurate predictions allow for scheduled maintenance before severe performance degradation or breakdowns occur. This minimizes unplanned downtime and ensures higher equipment availability.
- **Enhanced Production Scheduling and Resource Allocation:** Reliable OEE forecasts enable production managers to adjust schedules and resource deployment based on anticipated equipment performance changes. For instance, if the model predicts lower machine efficiency for the following day, managers can redistribute production loads or adjust work shifts accordingly. Conversely, during periods of high predicted OEE, operations can run at full capacity. This approach ensures optimal utilization of machinery and labor, contributing to resilient and efficient manufacturing systems aligned with Industry 4.0 objectives.

- **Handling Volatility and Noise:** One of the primary motivations for this study is the ability to capture complex, short-term patterns that traditional forecasting methods often overlook. Our approach is designed to handle the volatile nature of production environments more effectively. By incorporating noise-robust analysis techniques, we can smooth out random fluctuations while preserving meaningful deviations. This leads to more stable forecasts with fewer unexpected variations, ultimately improving confidence in production planning.
- **Competitive Advantage:** Accurate OEE forecasting provides manufacturers with a strategic advantage by ensuring consistently high equipment effectiveness. By leveraging predictive insights for proactive decision-making, companies can minimize unexpected losses, reduce operational costs, and meet production targets with greater reliability. In a competitive market, maintaining high OEE levels translates into better responsiveness to demand fluctuations, improved productivity, and overall enhanced manufacturing performance.

By addressing these challenges, our study aims to develop an advanced OEE forecasting approach that enhances predictive maintenance, optimizes production efficiency, and provides manufacturers with the insights needed to stay competitive in rapidly evolving industrial landscapes.

1.2 Literature Review

1.2.1 OEE

OEE was introduced as part of Total Productive Maintenance (TPM) to evaluate equipment efficiency [1]. Its calculation involves availability, which measures the proportion of scheduled time the equipment is available; performance, which assesses the equipment's operating speed compared to its designed speed; and quality, which evaluates the proportion of defect-free products. A perfect OEE score of 100% represents ideal manufacturing, where equipment operates at full speed without stoppages or defects [13]. OEE plays a critical role in manufacturing by highlighting inefficiencies in production [14], guiding continuous improvement initiatives [15], and enabling comparisons across production lines and facilities [13]. The ability to monitor and improve OEE ensures manufacturers remain competitive in dynamic markets [16]. The Fourth Industrial Revolution (4IR) has revolutionized OEE measurement and management by integrating advanced technologies such as the Internet of Things (IoT), machine learning, and big data analytics [17]. These innovations allow for real-time OEE monitoring and predictive analytics, providing manufacturers with actionable insights [18]. Examples include real-time monitoring through IoT-enabled sensors that provide continuous data on equipment performance [19], predictive analytics using machine learning models to forecast potential failures and OEE trends [20], and enhanced decision-making with advanced analytics optimizing resource allocation and maintenance schedules [17]. Traditionally, OEE is analyzed using statistical process control methods. This approach identifies losses and inefficiencies [16], such as downtime losses due to equipment failures, speed losses caused by suboptimal operating conditions, and quality losses resulting from defects. Studies have demonstrated the effectiveness of these methodologies in improving production performance [14]. With 4IR technologies, OEE management has transitioned from reactive to proactive strategies [18]. Advanced data analytics enables accurate forecasting, where predictive models estimate OEE trends, helping managers anticipate and address inefficiencies [20]. It also optimizes resource management by providing insights that improve team management and resource allocation [19], and reduces losses by targeting root causes of inefficiencies [17]. Forecasting OEE is essential in modern manufacturing for enhancing production planning and scheduling [18], minimizing downtime through proactive maintenance [20], and improving decision-making through accurate predictions [17]. Such advancements ensure manufacturing systems are resilient, efficient, and capable of meeting market demands. The integration of 4IR technologies into OEE management represents a paradigm shift in manufacturing. By leveraging advanced data analytics and IoT, manufacturers can not only monitor but also predict and optimize equipment performance. Future research should focus on refining predictive models and exploring their applications across diverse manufacturing sectors.

1.2.2 Forecasting

The evolution of forecasting methodologies has seen significant developments over the decades, beginning with traditional statistical models and advancing to complex ML and AI techniques. ARIMA models have long been a cornerstone in time series forecasting due to their ability to capture linear relationships in stationary datasets [9]. These models rely on identifying appropriate orders for autoregressive and moving average components, often complemented by differencing to ensure stationarity. Despite their strengths, ARIMA models face challenges when handling non-linear relationships and data with high volatility [21].

With the growing demand for accuracy in forecasting, particularly in industrial and financial domains, ML and AI models have emerged as robust alternatives. Methods such as support vector regression (SVR), random forests, gradient boosting machines (GBMs), and neural networks have demonstrated superior performance in capturing complex non-linear dependencies for long time series [22, 23]. The integration of exogenous variables, such as statistical features derived from historical data, has further enhanced the predictive capabilities of these models [10]. For instance, incorporating seasonality indices, moving averages, and lagged features has been shown to improve the accuracy of forecasts, particularly in scenarios involving external influences [24].

A recent frontier in forecasting involves the inclusion of topological features derived from time series data. These features, calculated using tools such as persistent homology, provide insights into the geometric and topological structure of data [25]. Topological Data Analysis (TDA) has proven particularly effective in handling volatile and short-term datasets, where traditional statistical and ML models may falter. The use of persistence diagrams, Betti curves, and persistence entropy allows models to capture intrinsic properties of data that are otherwise overlooked. For example, TDA-based features have been utilized in financial markets to predict stock price movements, demonstrating their utility in environments characterized by high uncertainty [26].

In conclusion, the integration of ARIMA, ML, and AI models, supplemented by exogenous variables, has significantly advanced the field of forecasting. Topological features, in particular, represent a promising avenue for enhancing predictions in volatile and short-term data scenarios, making them an essential component of modern forecasting techniques.

1.2.3 Topological Data Analysis

TDA has emerged as a powerful tool for analyzing complex and high-dimensional datasets, particularly in the presence of noise. One of its key advantages lies in its resilience to noise and its ability to extract meaningful patterns from the overall data structure. TDA operates on the premise that data has an inherent shape, and it utilizes this shape to identify topological features such as clusters, loops, and voids, which remain consistent despite perturbations in the data [27]. This approach has been successfully applied in multivariate time series studies, such as room occupancy data analysis, yielding improved prediction accuracy for classification tasks compared to traditional methods. The integration of TDA into time series forecasting pipelines offers promising advancements for achieving accurate and reliable predictions in dynamic environments, paving the way for future research and applications [28].

Recent advancements in time series forecasting have seen the integration of TDA techniques to capture complex temporal structures. Karan and Kaygun [29] developed a methodology that employs persistent homology to extract topological features from univariate time series. Using time-delay embeddings and windowing effectively reduced noise and enhanced classification accuracy on physiological datasets. Zeng et al. [30] introduced a novel topological attention approach, which integrates local topological features into attention mechanisms within forecasting models. This method demonstrated improved performance on benchmark datasets, highlighting the significance of topological structures in forecasting tasks. Han et al. [31] proposed a deep-learning model that combines TDA-derived features with temporal information to enhance intra-hour solar irradiance forecasting. Souto [32] introduced the concept of topological tail dependence, linking persistent homology with tail dependence theory to improve volatility forecasting in financial markets.

Despite these challenges, TDA's capacity to handle noise, analyze multiscale structures, and integrate with machine learning techniques has been demonstrated across domains including biomedical imaging, financial analytics, and intelligent transportation, making it a versatile tool for modern data analysis [33, 34].

The primary method of TDA, *Persistent Homology*, tracks the changes in these features across multiple scales, identifying persistent patterns more likely to represent significant underlying structures rather than noise [35]. Persistent homology visualizes these patterns using persistence diagrams or barcodes and allows for quantitative comparison through metrics such as bottleneck distance and Wasserstein distance.

Additionally, the Mapper algorithm, another key tool in TDA, provides a visual representation of data as graphs, enabling qualitative exploration of data structures and facilitating dimensionality reduction [36, 37]. However, the computational complexity of persistent homology, especially for large-scale or high-dimensional datasets, and the parameter dependency of Mapper highlight the ongoing challenges in the field [38].

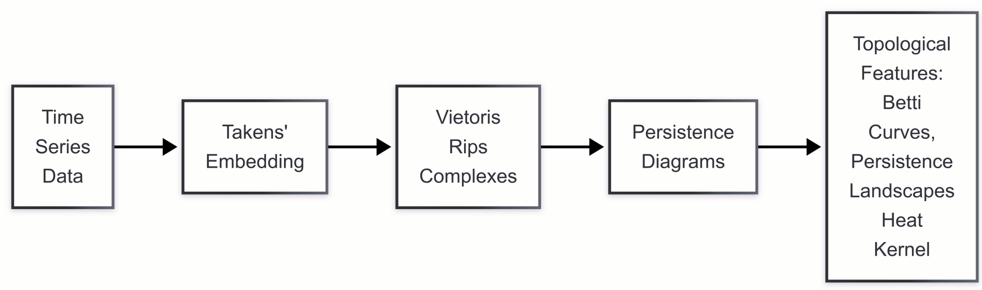


Fig. 1: Flowchart of the topological feature extraction

The TDA features extraction pipeline can be seen in Figure 1 [28]. Our study's primary strategy involves extracting robust features from time series data using persistent homology in line with the above pipeline. The process begins by transforming the original time series into a point cloud through Takens' embedding. This technique reconstructs the phase space of the dynamical system by mapping the time series into a higher-dimensional space, effectively capturing its temporal dynamics. Once the point cloud is obtained, we compute persistent homology using the Vietoris-Rips complex. This involves constructing a filtration nested sequence of simplicial complexes based on a chosen distance metric (commonly Euclidean). At each scale of the filtration, we analyze the topological components such as connected components (0-dimensional holes), loops (1-dimensional holes), and voids (2-dimensional holes). The birth and death of these components across scales are recorded in persistence diagrams, providing a multiscale summary of the data's topological structure. We convert the persistence diagrams into vectorised representations to integrate these topological insights into machine learning models. Giotto-TDA [39] offers several tools for this purpose, including persistence images, which transform persistence diagrams into fixed-size images by placing a Gaussian kernel on each point and summing the results, enabling the use of image-based machine learning techniques; Betti curves, which represent the number of topological components (Betti numbers) present at each scale, providing a concise summary of the data's topological complexity over the filtration; and persistence entropy, which quantifies the complexity of the persistence diagram by measuring the entropy of the lifespans of topological components, offering a single-value summary that can be used as a feature. These vectorized features are then utilized as inputs to our forecasting models, allowing us to capture and leverage the intrinsic topological structures within the time series data for improved predictive performance.

This study is a continuation of the research presented in [40], which investigated the forecasting of OEE in hydraulic press systems components vital to modern manufacturing plants' operational and financial performance. The previous work emphasized the challenges of forecasting short-term, volatile OEE data, which is often influenced by dynamic changes in production processes. To overcome the limitations of traditional forecasting models and standard artificial intelligence methods, that study introduced a novel approach by incorporating TDA into the forecasting pipeline. The model achieved more robust short-term predictions by leveraging topological features and provided actionable insights to enhance predictive maintenance and operational stability. Building on this foundation, the present study further advances the methodological framework and broadens its application across different production systems.

The remainder of this paper is structured as follows: Section 2 outlines the methodology, detailing the main steps involved. Section 3 provides a comprehensive explanation of the feature selection methods. Section 4 introduces the forecasting models. Section 5 presents the results, while Section 6 discusses the findings. Additionally, Section 7 gives the application of the proposed model in a manufacturing environment.

2 Methodology

With the advent of emerging technologies, many production plants now monitor OEE values in real-time. This capability enables the idea of forecasting OEE values, thereby facilitating the effective management of maintenance and production teams to improve the OEE values of the equipment. The proposed methodology integrates seamlessly with OEE monitoring systems and can be employed online to maintain or improve the stability of OEE trends.

In modern manufacturing plants, the technologies utilized in production equipment evolve rapidly, and continuous improvements are implemented to achieve higher efficiency levels. Consequently, forecasting models for OEE values must rely on short-term historical data to account for the dynamic nature of production environments.

Another critical aspect is the high volatility of OEE values, which arises from factors such as equipment breakdowns, setup changes, and quality-related issues. These challenges necessitate robust forecasting methodologies capable of handling the inherent variability in OEE data while providing actionable insights for operational decision-making.

Forecasting a time series with a short-term historical dataset and a volatile structure is a challenging task. To address this challenge, we propose a novel methodology designed specifically to handle these complexities.

2.1 Importance of the Proposed Model

1. **Robust Forecasting with Short-Term, Volatile Data:** The proposed model is designed to generate reliable forecasts using short-term datasets characterized by high volatility, a common challenge in dynamic production environments.
2. **Practical Applicability in Real-World Manufacturing:** The forecasting accuracy achieved by the model enables informed decision-making and efficient resource management in production facilities.
3. **Enhanced Forecasting Through Time Series Decomposition:** By decomposing time series data into trend, seasonal, and residual components, the model improves forecasting accuracy by effectively capturing distinct patterns in each component.
4. **Optimized Feature Extraction via Topological Data Analysis:** The integration of TDA enables the extraction of meaningful geometric features, while a rigorous feature selection process ensures that only the most relevant information contributes to improved predictive performance.

5. **Improved Predictive Maintenance and Operational Efficiency:** Accurate short-term forecasts, combined with robust feature extraction, support proactive maintenance strategies, minimizing unplanned downtime and enhancing overall operational efficiency.

2.2 Description of Our Methodology

The methodology begins with short term hourly OEE data that exhibit fluctuations and high volatility, stemming from dynamic production processes. To address this complexity, the time series is decomposed into three distinct components: the trend, seasonal components, and residuals. This decomposition enables targeted modeling of each component, improving forecasting accuracy and robustness. Figure 2 provides a visual summary of the methodology.

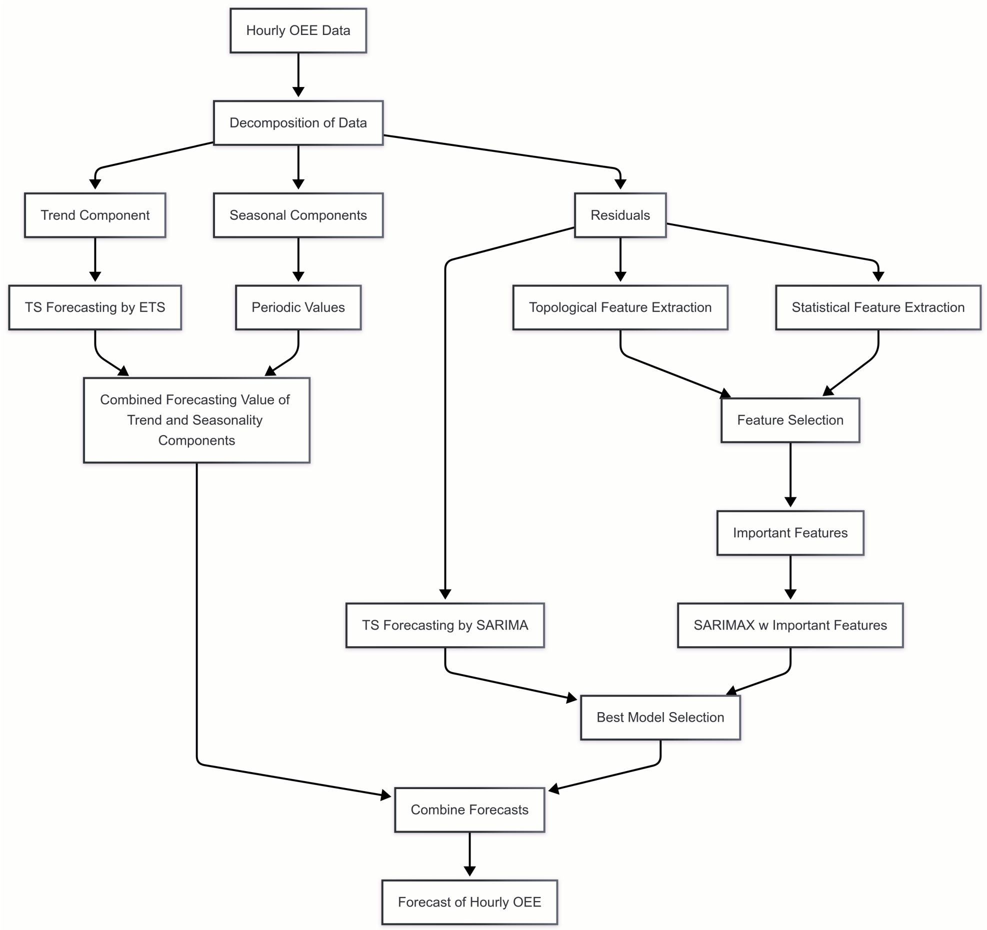


Fig. 2: Flowchart of the proposed method

The trend component captures the overall movement in OEE values and is forecasted using the Exponential Smoothing State Space Model (ETS), which effectively models gradual changes over time. Seasonal components represent periodic patterns due to production cycles, such as 8-hour shifts, daily (24-hour), and weekly (168-hour) cycles. These components are modeled and forecasted as periodic values, reflecting recurring production patterns. The residuals, which represent irregular variations and noise not captured by the trend or seasonality, are forecasted with S/ARIMA. The residual forecasting is further enhanced with features derived from TDA and statistical measures, enabling the models to capture intricate and volatile patterns.

The extracted feature set is extensive, making feature selection a critical step in the methodology. We applied a well-defined approach, which combines Recursive Feature Elimination (RFE) and Particle Swarm Optimization (PSO).

Using the extracted features, the proposed model employs a SARIMAX approach for forecasting. The model integrates statistical and topological features as exogenous variables to enhance predictive accuracy. By leveraging these additional informative inputs, the SARIMAX model effectively captures both temporal dependencies and complex feature interactions, improving overall forecast performance. The trend and seasonal components have been individually forecasted, and their outputs are then integrated with the forecasted residual component derived from the decomposition process, providing a comprehensive representation of the overall OEE pattern.

The outputs of forecasting models are evaluated through the best model selection process, which identifies the optimal approach based on predefined performance criteria such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). This iterative evaluation ensures that the most accurate and robust forecasting method is selected.

Additionally, a benchmark set is established by applying classical forecasting approaches alongside recent transformer-based foundation models. This benchmarking allows for a comparative evaluation, assessing the effectiveness of the proposed SARIMAX model against both traditional statistical methods and state-of-the-art deep learning architectures.

We will further elaborate on the application of the proposed model across different datasets.

2.3 Data

In this study, we utilize three datasets collected from different equipment, labeled as GM2, H2, and GH2. GH2 and GM2 are advanced, highly complex system that manufacture the inner body group of standard household appliances, while H2 is hydraulic press system with four presses dedicated to producing stainless steel body parts. The equipment can be seen in Figure 3 and 4.

To evaluate the generality of the proposed model across different types of equipment, we apply it to these three distinct systems, each exhibiting unique operational characteristics and complexities. This assessment ensures the robustness and adaptability of the model in diverse manufacturing environments.

Each dataset represents hourly OEE values, ranging between 1 and 60, where 1 indicates no operation, and 60 indicates full operation. The datasets exhibit high

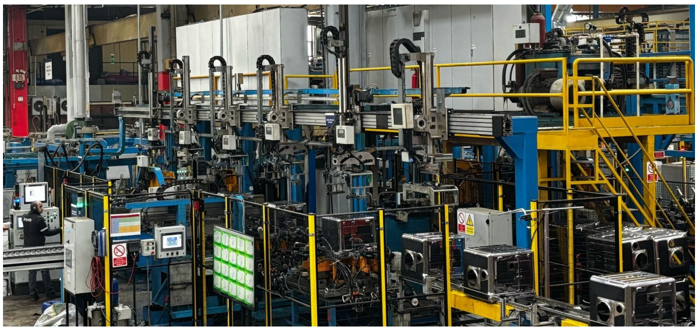


Fig. 3: Production Equipment GH2

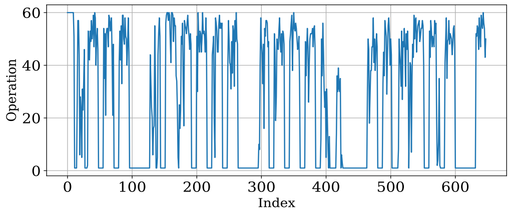
Fig. 4: Production equipment H2

variability, as indicated by the standard deviation values, and include a wide range of operational states from the minimum to the maximum. Table 1 is a summary of the statistical properties of the datasets.

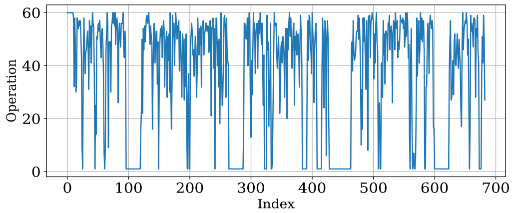
Table 1: Statistical Summary of the Datasets

Statistic	GH2 Operation	H2 Operation	GM2 Operation
Count	648	683	672
Mean	26.58	36.44	30.30
Std Dev	24.19	21.90	24.86
Min	1.00	1.00	1.00
25th Percentile	1.00	16.00	1.00
Median (50%)	30.00	46.00	41.00
75th Percentile	51.00	54.00	54.00
Max	60.00	60.00	60.00

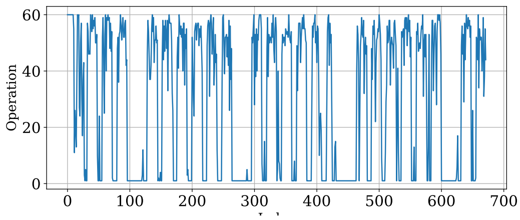
As one can see, the datasets consist of approximately 680 hours of data, representing short-term observations over a period of approximately one month. The standard deviation across the datasets is at least 20.72, indicating high volatility within the data range of 1 to 60. This variability reflects the dynamic nature of equipment operation, emphasizing the need for robust forecasting methodologies to manage such fluctuations effectively. The time series plots of the datasets are given in Figure 5.



(a) GH2



(b) H2



(c) GM2

Fig. 5: Time series plots of hourly OEE values (1-60) for GH2, H2, and GM2 datasets.

In Figure 6, the histogram of OEE values for equipment GM2 is shown. Normality tests including the Shapiro-Wilk test, D'Agostino and Pearson's test, and the Anderson-Darling test indicated that the time series deviates from normality. Furthermore, the skewness and kurtosis were calculated as -0.69 and -1.04, respectively, underscoring the need for forecasting methodologies that can effectively handle non-normal distributions.

In the time series plots of the datasets, a complex structure of seasonality is evident. From a domain expert perspective, the datasets are assumed to exhibit multiple

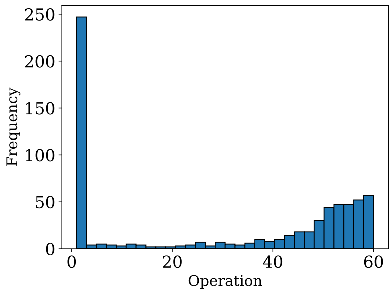
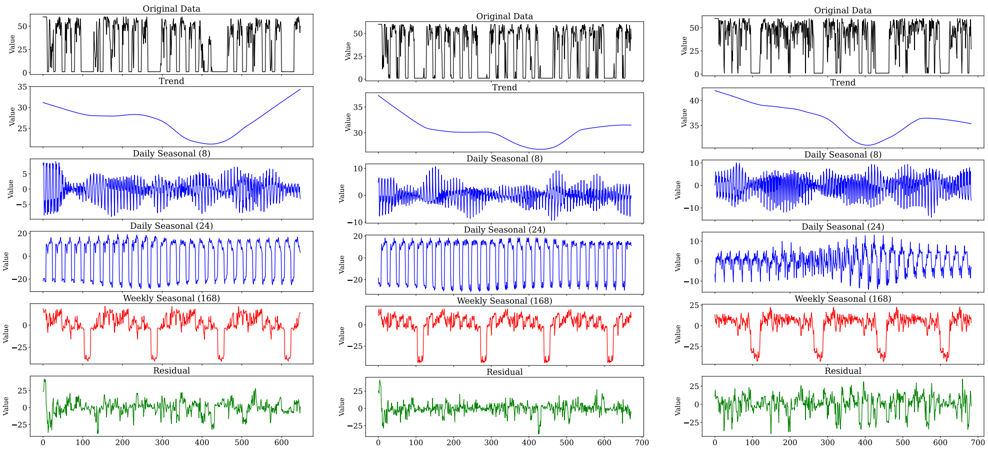


Fig. 6: Histogram of OEE values of equipment GM2

seasonalities. The possible seasonality values include 8 hours (shift duration), 24 hours (daily cycles), and 168 hours (weekly cycles). These periodic patterns are crucial for understanding the operational characteristics of the equipment. The decomposition plot of the datasets, illustrating these seasonal components, is shown in Figures 7.



(a) GH2 (b) GM2 (c) H2

Fig. 7: Time series decomposition of GH2, GM2, and H2 datasets.

After decomposition, the residual component of the OEE time series data as depicted in bottom plots of Figure 7 approximates a normal distribution, representing a significant improvement. The decomposition approach effectively mitigates the original non-normality of the data.

In time series forecasting, stationarity is a critical requirement. To ensure this, we evaluated the need for both regular and seasonal differencing. The Kwiatkowski-Phillips-Schmidt-Shin (KPSS), Osborn-Chui-Smith-Birchenhall (OCSB), and Canova-Hansen (CH) tests confirmed that the series are stationary in both respects, a validation that is essential for effectively applying time series models and achieving reliable forecasts.

Figure 8, as a representative example of three datasets, displays the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) graphs of the residual component as derived for the SARIMA model.

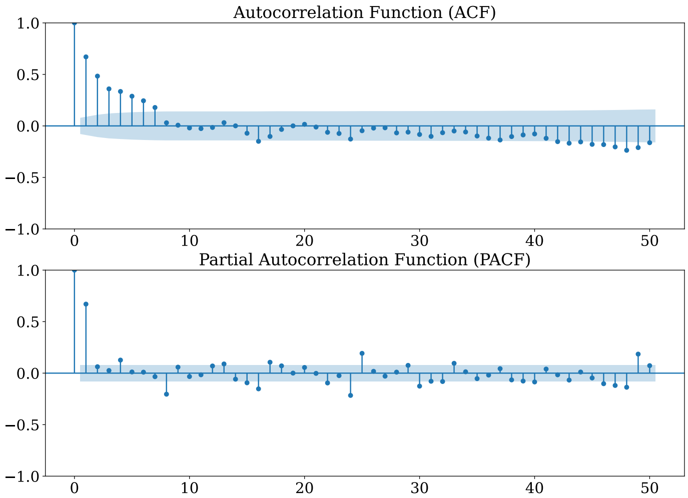


Fig. 8: GH2 dataset

Forecasting methods typically require the incorporation of exogenous variables to enhance model performance. These variables are extracted using traditional statistical techniques as well as innovative approaches like topological data analysis (TDA). In the following sections, we delve deeper into the feature extraction process to generate the exogenous variables used in SARIMAX.

2.4 Statistical Feature Extraction

Using the `tsfresh` library, a comprehensive set of statistical features is extracted from the residual component of OEE time series data [41].

- **Descriptive and Deviation Metrics:** This group includes basic statistical measures that describe the central tendency and dispersion of the data, such as the sum, mean, median, standard deviation, variance, skewness, kurtosis, and root mean square. In addition, features like absolute energy, mean absolute change, and mean second derivative central capture the magnitude of deviations and the rate of change within the series.[42, 43]
- **Frequency Domain Features:** By applying Fourier analysis, frequency domain characteristics are derived. This category comprises Fourier coefficients (including their real, imaginary, absolute, and angular components) and aggregated statistics

such as centroid, variance, skewness, and kurtosis. These features help to reveal dominant frequencies and periodic patterns in the data [44, 45].

- **Autocorrelation and Temporal Dependency Measures:** To quantify the persistence and repeating patterns in the time series, autocorrelation and partial autocorrelation features are computed over multiple lags. Aggregated autocorrelation metrics provide an overall assessment of the temporal dependencies within the data [42–44].
- **Entropy and Complexity Measures:** Entropy-based features, including sample entropy, approximate entropy, permutation entropy, and Fourier entropy, are extracted to assess the complexity and randomness inherent in the time series. These measures are crucial for evaluating the predictability of the data [45, 46].
- **Trend and Change Quantile Analysis:** Additional features capture the linear trend characteristics (e.g., slope, intercept, and associated statistics) as well as change quantile metrics, which summarize how the signal shifts over time. These features aid in identifying systematic changes and trends in the series [44].

2.5 Topological Feature Extraction

The proposed methodology leverages TDA to extract a rich set of features from time series data, thereby enhancing forecasting accuracy and supporting advanced analytical applications. This section details both the overall extraction process and the underlying principles of its key components[47] [48].

Initially, the raw time series is converted into an array and segmented into overlapping windows using a sliding window transformation (with a window size of 24 and a stride of 1). This segmentation preserves local temporal dynamics by isolating short-term patterns within each window [49]. Each window is then embedded into a higher-dimensional space through Takens Embedding [50], which reconstructs the phase-space trajectories based on a specified time delay (8) and embedding dimension (3). The selection of the time delay, denoted as τ, is critical; it ensures that the successive components of the embedding vector are sufficiently independent, thus faithfully unfolding the dynamics of the original system without introducing redundancy [51] .

The time delay τ is typically determined using either the autocorrelation function choosing the delay at the first zero crossing or when the function decays to $1/e$ of its initial value or via the average mutual information method by selecting the first local minimum [49]. In contrast, the embedding dimension m is commonly estimated using the False Nearest Neighbors algorithm, which identifies the minimal dimension where false neighbors are minimized, or by applying Cao's method to detect saturation in dynamical invariants [52] . For each sliding window, the time delay and embedding dimensions are calculated and the representative ones are selected [28] .

Following the embedding, the VietorisRips Persistence method is applied to the transformed data. This technique constructs a simplicial complex by connecting points within a chosen distance threshold, and as the threshold varies, it tracks the appearance and disappearance of topological features such as connected components, loops, and voids across different scales [53], [49]. The outcome is a persistence diagram that records the birth and death of these features across homology dimensions H_0, and H_1

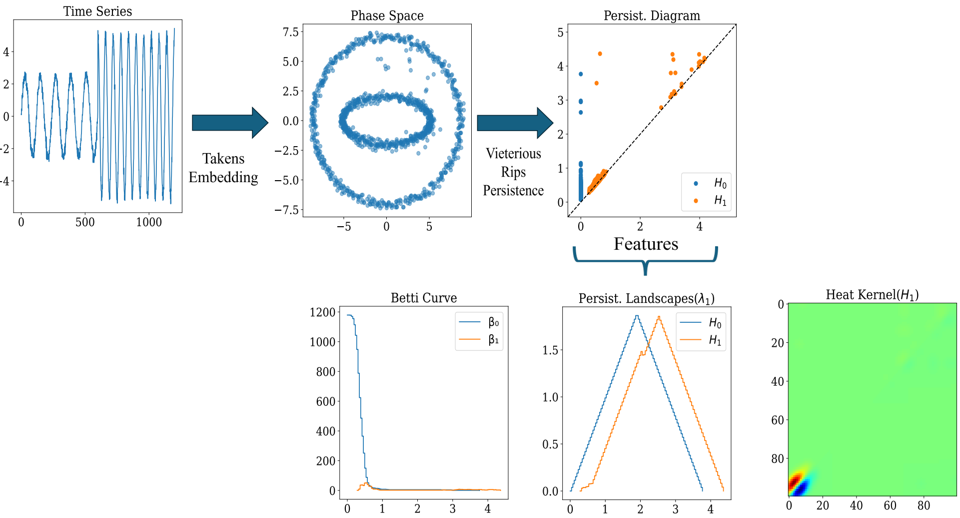


Fig. 9: The workflow of TDA feature extraction

[51] . To ensure consistency and comparability across different data samples, the persistence diagrams are scaled appropriately. The standard feature extraction steps for TDA can be seen in Figure 9 [54] [52] [55] .

A variety of TDA based feature extractors are then employed on the scaled persistence diagrams. These extractors capture different aspects of the topological signatures in the data:

- **Persistence Entropy:** This extractor computes the Shannon entropy of the lifetimes (i.e., the differences between death and birth times) of topological features, quantifying the overall complexity and disorder in the persistence diagram. Higher entropy values indicate a more uniform distribution of lifetimes, while lower values suggest dominance by a few persistent features [56] [57] [49] [55].
  Let D be a persistence diagram consisting of a set of persistence pairs (b_i, d_i), where b_i is the birth time and d_i is the death time of a topological feature.

  1. **Compute Lifetimes:** The lifetime of each feature is given by:

  $$l_i = d_i - b_i \tag{1}$$

  where l_i represents the persistence of the feature.

2. **Compute Probability Distribution:** Define the probability mass function:

$$p_i = \frac{l_i}{\sum_j l_j} \tag{2}$$

which normalizes the lifetimes to form a probability distribution.

3. **Compute Shannon Entropy:** The persistence entropy is given by:

$$H = -\sum_i p_i \log p_i \tag{3}$$

where the logarithm is typically taken in base e (natural logarithm).

- **Amplitude Metrics:** These features quantify the "amplitude" of the persistence diagram by measuring distances or functional summaries between topological features[58] . In particular:
  - **Bottleneck Amplitude:** Uses the bottleneck distance to assess the maximum difference between matched features, thus focusing on the most significant topological discrepancies [59].
    The *bottleneck distance* between two persistence diagrams D_1 and D_2 is defined as:

    $$d_B(D_1, D_2) = \inf_{\gamma} \sup_i \|(b_i, d_i) - \gamma(b_i, d_i)\|_\infty \tag{4}$$

    where γ ranges over all bijections between the two diagrams and $\| \cdot \|_\infty$ is the supremum norm.
  - **Wasserstein Amplitude:** Aggregates differences over all matched features to provide a global similarity measure between persistence diagrams[58] .
    The *Wasserstein distance* of order p is given by:

    $$d_W(D_1, D_2) = \left(\inf_{\gamma} \sum_i \|(b_i, d_i) - \gamma(b_i, d_i)\|^p \right)^{\frac{1}{p}} \tag{5}$$

    where the exponent p controls how differences are aggregated [52].

- **Betti Curves and Persistence Landscapes:**
  - **Betti Curves:** These plot the number of topological features (i.e., Betti numbers) as a function of the filtration parameter, illustrating how the counts of connected components, loops, or voids evolve over scales[58] [60]. Betti curves describe the evolution of topological features by tracking the Betti numbers $\beta_n(t)$ over a filtration parameter t. The Betti number β_n counts the number of n-dimensional topological features, such as:

    $$\beta_0(t) = \text{Number of connected components at filtration level } t, \tag{6}$$
    $$\beta_1(t) = \text{Number of loops (1-dimensional cycles) at filtration level } t, \tag{7}$$
    $$\beta_2(t) = \text{Number of voids (2-dimensional cavities) at filtration level } t. \tag{8}$$

The Betti curve is then given by:

$$\mathcal{B}_n(t) = \sum_{(b_i,d_i)\in D_n} \mathbf{1}_{[b_i,d_i)}(t) \tag{9}$$

where $\mathbf{1}_{[b_i,d_i)}(t)$ is an indicator function that is 1 if t is within the lifespan of the feature and 0 otherwise [52] [49].

– **Persistence Landscapes:** These provide continuous functional summaries of persistence diagrams by mapping them into a collection of piecewise linear functions . They offer a stable representation that facilitates statistical analysis and integration with machine learning methods [61]. Persistence landscapes transform persistence diagrams into piecewise linear functions, allowing statistical analysis. The persistence landscape function $\lambda_k(t)$ is defined as:

$$\lambda_k(t) = \sup_{\substack{(b_i,d_i)\in D \\ k\text{-th largest}}} \max(0, \min(t - b_i, d_i - t)). \tag{10}$$

The persistence landscape norm is computed using an L^p-norm:

$$L_p = \left(\int_{-\infty}^{\infty} \sum_k |\lambda_k(t)|^p dt \right)^{\frac{1}{p}}. \tag{11}$$

- **Silhouette and Heat Kernel Representations:** These representations offer alternative continuous summaries of persistence diagrams.
  – **Silhouette Representation:** Computes a weighted average of the persistence landscape, emphasizing the most significant features while reducing noise. This silhouette effectively captures the quality of the topological signature [62] [60] . The silhouette function provides a weighted summary of the persistence landscape, reducing noise while emphasizing significant features. It is defined as:

$$S_\alpha(t) = \frac{\sum_{(b_i,d_i)\in D} (d_i - b_i)^\alpha \lambda_1(t)}{\sum_{(b_i,d_i)\in D} (d_i - b_i)^\alpha} \tag{12}$$

where $\alpha \geq 0$ is a weighting parameter that controls the contribution of each persistence feature. A higher α emphasizes longer-persisting features [49] [60] .

  – **Heat Kernel Representation:** Smooths the discrete persistence diagram using a heat diffusion process, yielding a continuous function that encapsulates the global structure of the topological features. This representation is particularly useful for downstream machine learning tasks. The heat kernel representation smooths the persistence diagram using a Gaussian heat diffusion process, transforming the discrete persistence pairs into a continuous function [63] . It is defined as:

$$H(t,\sigma) = \sum_{(b_i,d_i)\in D} \frac{1}{\sqrt{4\pi\sigma^2}} e^{-\frac{(t - \frac{b_i + d_i}{2})^2}{4\sigma^2}} \tag{13}$$

where σ controls the diffusion scale.

Each of these extractors produces a feature matrix that is subsequently flattened and organized into a tabular format. In addition to these TDA features, statistical measures such as the lifetimes of features and various summary statistics (total, mean, median, variance, standard deviation, maximum, and minimum) are computed for each homology dimension to further enhance the descriptive power of the dataset[48] [55] .

Finally, all extracted features (including both the TDA descriptors and their statistical summaries) are concatenated into a single, comprehensive feature set. To improve model efficiency and reduce redundancy, feature selection techniques such as low variance elimination and correlation-based filtering are applied. The final refined TDA feature set provides a robust, multi-faceted representation of the time series, capturing both its dynamic behavior and its underlying topological structure, and thereby supporting more accurate forecasting and advanced analytical applications[47] [55] .

3 Feature Selection

In the feature extraction phase, the initial feature set exceeds 450 dimensions. Such high dimensionality not only introduces the well-known *curse of dimensionality* but also hampers the forecasting performance of our models.

To address this, feature selection begins with a low variance elimination step, where features with variance below a threshold of 0.01 are removed. This filtering ensures that only the most informative and discriminative features are retained for use as exogenous variables in forecasting models such as SARIMAX.

Afterward, the primary challenge becomes multicollinearity. By identifying and eliminating dependent columns, the number of features is further reduced. This step prevents redundancy and ensures that retained features provide unique and valuable information.

To further refine feature selection, a smart selection process is implemented to improve machine learning model performance. The idea is to detect and remove highly correlated features, as multiple strongly correlated variables can lead to redundancy and overfitting. Instead of arbitrarily dropping one of the correlated features, their importance is evaluated using a machine learning model Random Forest in this case. The method selects the most useful features based on their contribution to predicting the target variable.

After these steps, we reduced the number of features to approximately 150 for both topological and statistical feature sets. However, this is still a high dimensionality that may negatively impact forecasting performance.

To further refine the feature set and retain only those features that significantly contribute to forecasting, we design feature selection strategies that align with our objectives. These strategies aim to enhance model interpretability, improve computational efficiency, and optimize predictive performance by systematically selecting the most relevant features.

3.1 Feature Selection via SARIMAX Model Performances

Our proposed method for feature selection is based on statistical significance and the Bayesian Information Criterion (BIC). It consists of two steps. In the first step, features are recursively selected based on their p-values to retain only statistically significant predictors. If further refinement is necessary, an additional feature selection step is performed using PSO to optimize the final set of features, ensuring both statistical relevance and predictive performance. The flowchart illustrating the feature selection methods is shown in Figure 10.

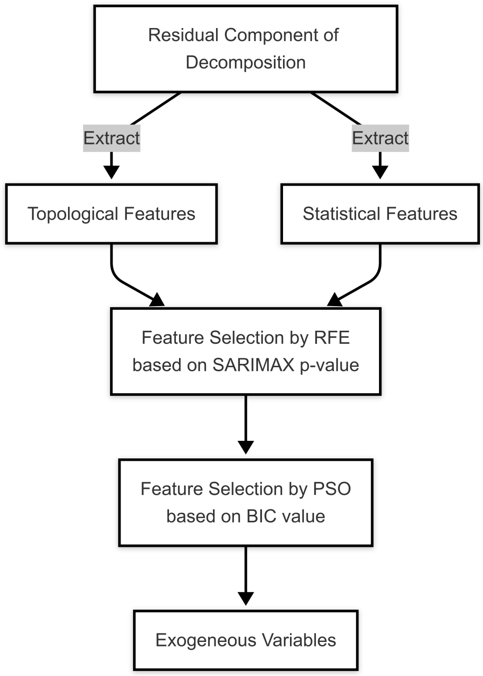


Fig. 10: Flowchart of the feature selection methods

3.1.1 First Step: Recursive Feature Elimination via p-values

A SARIMAX model is fitted using the available features, and the p-values for the exogenous predictors are computed. Features that exhibit p-values greater than a predefined significance level (e.g., 0.05) are considered statistically insignificant.

This procedure is implemented in a recursive fashion. At each iteration, the SARIMAX model is refitted, and the p-values for all current features are examined. A subset of features with p-values exceeding 0.05 is identified. The least significant features are removed, and the model is refitted on the reduced set. This process repeats until fewer than a set number of features (or none) have p-values above the threshold. This recursive elimination ensures that only those features that have a statistically significant impact on the target variable are retained.

3.1.2 Second Step: Feature Elimination by PSO via BIC

In cases where the feature set remains excessively large after applying the RFE method, we further refine the selection using PSO. PSO is a population-based stochastic optimization technique inspired by the social behavior observed in flocks of birds or schools of fish. In our framework, each particle in the swarm represents a candidate subset of features, encoded as a binary vector indicating selection. The PSO algorithm iteratively updates the swarm by evaluating the performance of each feature subset using the BIC of a SARIMAX model fitted on that subset. The objective is to minimize the BIC, which serves as a measure of model quality that penalizes complexity. By doing so, the PSO method effectively searches the space of possible feature combinations to identify a parsimonious subset that leads to superior forecasting performance.

To ensure a well-optimized selection process, we configure the PSO parameters as follows: a swarm size of 40 particles, a maximum of 300 iterations, an inertia weight of $\omega = 0.7$ to balance global and local search, and cognitive and social parameters set to $\phi_p = 1.4$ and $\phi_g = 1.8$, respectively, to regulate individual and swarm-wide best solutions. The optimization is executed five times, with BIC values recorded for each run. At the end of these iterations, two feature selection strategies are applied: (i) the feature subset corresponding to the lowest BIC value across all runs is selected as *Set 1*, representing the most optimal feature set; (ii) a secondary feature set is constructed by selecting features that appeared in at least three out of the five runs, ensuring stability and consistency in feature importance.

By implementing this hybrid feature selection framework, we significantly reduce dimensionality while preserving the most informative and discriminative features, ultimately enhancing model interpretability and forecasting performance. The number of extracted and selected features can be seen in Table 2.

Table 2: Feature Summary for GH2, H2, and GM2 Datasets

	GH2 Dataset	**H2 Dataset**	**GM2 Dataset**
Statistical Features	267	268	264
RFE Elimination	57	48	37
PSO Elimination	8	11	14
Topological Features	144	143	131
RFE Elimination	19	55	52
PSO Elimination	3	5	6

4 Forecasting Models

4.1 Benchmark Model Set:

In forecasting step, we first define a benchmark model set. Start with a SARIMA model applied to the given datasets GH2, H2 and GM2 without decomposition. The benchmark forecasting models considered include classical approaches such as ETS, and TBATS, as well as recent transformer-based foundation models such as CHRONOS, TimesFM, and Lag-Llama.

4.1.1 Exponential Smoothing (ETS)

ETS stands for *Error*, *Trend*, and *Seasonality*, referring to a family of exponential smoothing methods. The ETS framework models the data-generating process by combining three main components. The Error (E) represents the type of error process, which can be additive or multiplicative. The Trend (T) describes whether there is no trend, an additive trend, or a multiplicative trend. The Seasonality (S) indicates whether there is no seasonality, additive seasonality, or multiplicative seasonality [64] . These components are combined in different ways to accommodate a wide range of time series patterns. Because ETS relies on exponential smoothing techniques, it can be highly effective for short-term forecasts and is often robust in the presence of noise.

4.1.2 TBATS Model

The TBATS model is designed to handle complex seasonality and non-integer (potentially multiple) seasonal periods. The name is an acronym representing different components of the model. The T refers to trigonometric terms used for modeling seasonality. The B denotes the Box-Cox transformation, which stabilizes variance. The A represents ARMA errors (AutoRegressive Moving Average) to capture additional temporal dependencies. The second T accounts for the trend component, accommodating level or slope changes. The S corresponds to seasonal components that may be non-integer or multiple in frequency [65]. By combining these components, TBATS can effectively capture highly irregular seasonal patterns and is especially useful for time series with complex seasonalities, such as daily data with weekly and yearly effects.

4.1.3 Recent Advancements in Time Series Forecasting Models

Recent research has focused on harnessing foundation models large-scale transformer architectures initially developed for natural language processing to capture intricate temporal dependencies. Key developments include:

- **Chronos:** Chronos transforms continuous time series data into discrete token sequences through scaling and quantization. It then leverages transformer-based language models (particularly T5 variants) trained with a cross-entropy loss. Pre-trained on a diverse set of public and synthetic datasets (via Gaussian processes), Chronos exhibits strong zero-shot performance and adapts well with fine-tuning, thereby simplifying the forecasting pipeline while achieving competitive accuracy [66] .

- **Lag-Llama:** Lag-Llama is the first open-source foundation model tailored for univariate probabilistic time series forecasting. Employing a decoder-only transformer architecture that incorporates lagged covariates, it effectively captures long-range dependencies. Trained on a vast, diverse corpus of time series data, Lag-Llama demonstrates robust zero-shot generalization and often outperforms conventional deep learning approaches when fine-tuned on smaller datasets [67].
- **TimesFM:** Developed by Google, TimesFM is a rapid, decoder-only forecasting model designed for out-of-the-box predictions. With a streamlined architecture and minimal preprocessing, TimesFM delivers swift and accurate forecasts, making it particularly well-suited for real-time applications where prediction speed is paramount [68] .

4.2 Proposed Model

Our proposed model builds on the strengths of SARIMAX model and featuring power of TDA. We start by decomposing each time series into three distinct components: seasonal, trend, and residual. The seasonal and trend components are forecast using standard forecasting models, while the residual component is modeled via a SARIMAX framework that incorporates exogenous variables. The methodology is depicted in Figure 11. Exogenous variables are extracted from each window of size 24. To prevent data leakage, the forecasted value is always the next one, utilizing exogenous variables derived from the previous 24 time series values [51].

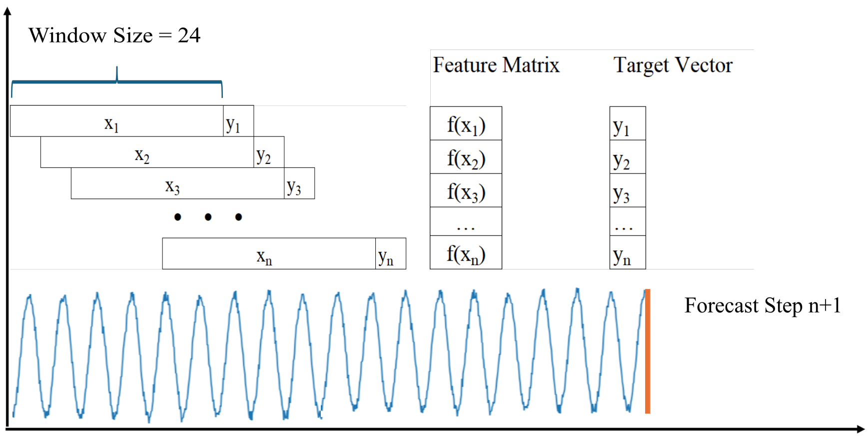


Fig. 11: The forecasting strategy for our proposed model

We extract both topological and statistical features from the data. A rigorous feature elimination process is then applied to retain only the most informative predictors. This two-stage strategy enables our model to capture both over all trends and short-term irregularities, thereby enhancing overall forecasting accuracy.

4.2.1 SARIMAX Modeling

Our baseline method employs SARIMAX models, which have long been a standard approach in time series analysis. To determine the appropriate model orders, we analyze the ACF and PACF plots of the residual components obtained from the time series decomposition. Based on these plots, the optimal autoregressive (AR) and moving average (MA) orders are selected to capture temporal dependencies effectively.

- For dataset GH2, we select a non-seasonal order of $(4, 0, 0)$ combined with a seasonal order of $(1, 0, 1)$ with a seasonal period of 8, yielding the model:

$$\text{SARIMAX}(4, 0, 0)(1, 0, 1)_8.$$

- For dataset GM2, the non-seasonal order is adjusted to $(2, 0, 0)$ while maintaining the same seasonal specification, resulting in:

$$\text{SARIMAX}(2, 0, 0)(2, 0, 1)_8.$$

- For dataset H2, the non-seasonal order is adjusted to $(1, 0, 0)$ while maintaining the same seasonal specification, resulting in:

$$\text{SARIMAX}(1, 0, 0)(1, 0, 1)_8.$$

5 Results

The results of the benchmark models are presented in Table 3. The models MAE, MAPE and Cost values are given. The cost values denote the operating time of the algorithms on a standard Intel i7 machine. Model performances are evaluated using MAE and MAPE metrics.

In this benchmark group, no decomposition was applied to the OEE datasets. Across all datasets, recent time series models Chronos, Lag-Llama, and TimesFM failed to outperform traditional models ETS, TBATS, and SARIMA. In each dataset, at least one of the traditional models achieved superior performance compared to the newer approaches.

Tables 4, 5, 6 present the performance of our proposed SARIMAX models across the GH2, H2, and GM2 datasets, utilizing topological or statistical features as exogenous variables. In our forecasting strategy, we first evaluate the impact of incorporating exogenous variables either statistical or topological into the baseline SARIMA model. If the inclusion of these variables leads to improved forecasting performance, we proceed without applying any feature selection, thereby avoiding the additional computational cost. However, in cases where this approach does not yield satisfactory

Table 3: Comparison of Forecasting Models across no decomposition GH2, H2, and GM2 Datasets

Dataset	GH2			H2			GM2		
Model	**MAE**	**MAPE**	**Cost**	**MAE**	**MAPE**	**Cost**	**MAE**	**MAPE**	**Cost**
CHRONOS	4.56	0.10	10	12.50	0.32	9	12.10	0.34	9
Lag-Llama	4.76	0.10	8	14.32	0.31	9	11.84	0.33	9
TimesFM	7.37	0.14	7	23.64	0.48	7	14.28	0.40	10
ETS	14.49	0.28	5	8.00	0.19	6	9.42	0.17	6
TBATS	18.43	0.36	38	9.80	0.22	40	8.87	0.17	42
SARIMA	3.81	0.08	5	14.11	0.29	33	11.26	0.31	4

results, we introduce a feature selection procedure to refine the input set. To maintain clarity and reduce complexity, we omit the intermediate results from this stepwise process and report only the final outcomes that reflect the most effective modeling configuration.

Table 4: Performance of SARIMAX models on GH2 dataset with and without feature selection and different exogenous inputs

Model	Feature Selection	Exogenous	MAE	MAPE	Cost
SARIMA	None	None	7.36	0.15	3
SARIMAX	No	Topological	**3.14**	**0.07**	80
SARIMAX	Yes (RFE, PSO)	Topological	–	–	–
SARIMAX	No	Statistical	–	–	–
SARIMAX	Yes (RFE, PSO)	Statistical	4.96	0.10	1016

Table 5: Performance of SARIMAX models on H2 dataset with and without feature selection and different exogenous inputs

Model	Feature Selection	Exogenous	MAE	MAPE	Cost
SARIMA	None	None	3.88	0.10	7
SARIMAX	No	Topological	–	–	–
SARIMAX	Yes (RFE, PSO)	Topological	**3.59**	**0.10**	520
SARIMAX	No	Statistical	–	–	–
SARIMAX	Yes (RFE, PSO)	Statistical	5.35	0.13	1031

We also explored the combination of topological and statistical features as exogenous inputs in the SARIMAX models. However, this integrated approach did not lead

Table 6: Performance of SARIMAX models on GM2 dataset with and without feature selection and different exogenous inputs

Model	Feature Selection	Exogenous	MAE	MAPE	Cost
SARIMA	None	None	7.84	0.21	3
SARIMAX	No	Topological	**4.74**	**0.14**	106
SARIMAX	Yes (RFE, PSO)	Topological	–	–	–
SARIMAX	No	Statistical	8.61	0.24	157
SARIMAX	Yes (RFE, PSO)	Statistical	–	–	–

to any significant improvement in forecasting accuracy compared to using each feature type individually. As a result, we did not pursue this strategy further in our final model configurations.

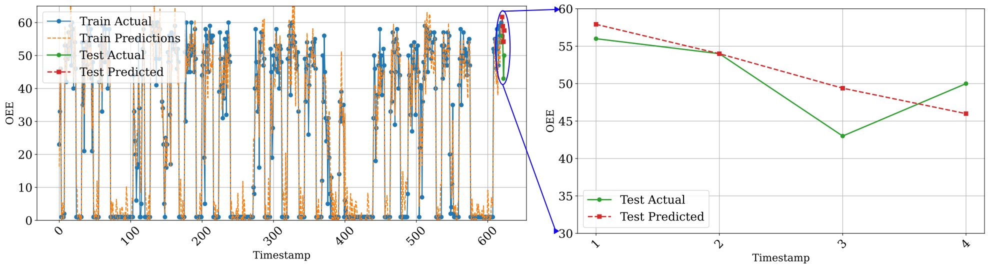


Fig. 12: The forecasting result of the proposed model for dataset GH2

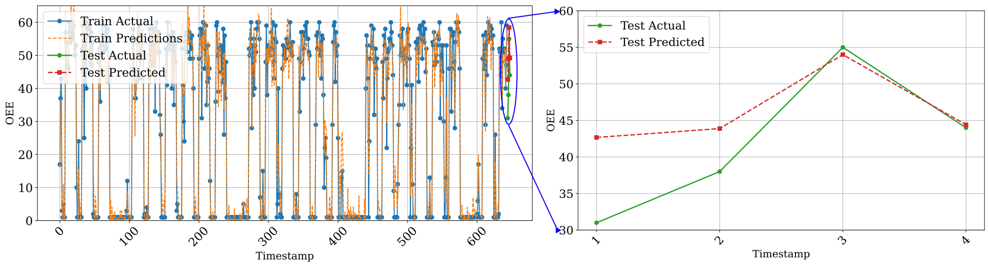


Fig. 13: The forecasting result of the proposed model for dataset GM2

This approach consistently outperforms all benchmark models, achieving at least a 17% improvement in both MAE and MAPE. The most notable results are observed when topological features are used without additional feature selection, particularly in the GH2 and GM2 datasets.

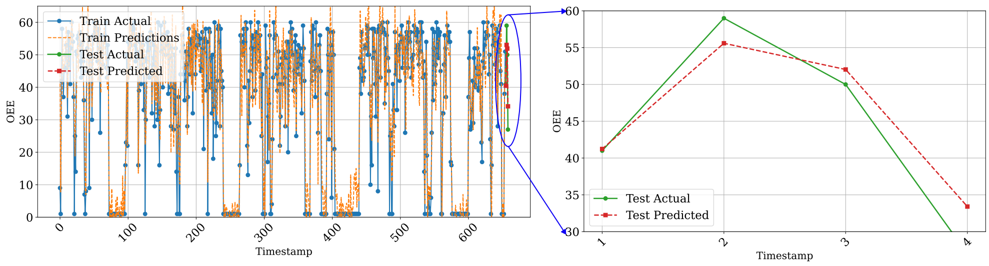


Fig. 14: The forecasting result of the proposed model for dataset H2

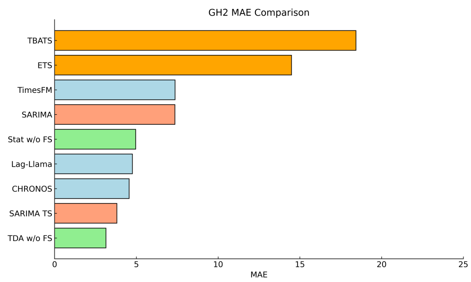
However, for the H2 dataset, feature selection is necessary to enhance predictive accuracy, indicating that the effectiveness of topological features can be dataset-specific. The feature selection algorithms used are RFE and PSO, as described above. The SARIMAX model, enhanced with topological features as exogenous variables and optimized through RFE and PSO, delivered the best results when applied to the residual component of the time series decomposition of H2. Since testing all possible feature combinations is an NP-hard problem, PSO was chosen as a heuristic feature selection method. Due to the stochastic nature of PSO, the algorithm was executed 15 times, and the features most frequently selected were compiled into the PSO dataset.

This configuration outperformed all other models and feature selection methodologies for H2. Finally, all results obtained are considered acceptable according to both domain expert evaluations and standard practices in time series forecasting.

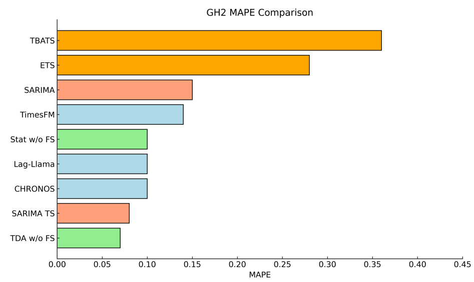
Moreover, the computational cost of our proposed model remains within an affordable range. Notably, the SARIMAX model using topological features without feature selection yields competitive performance while maintaining a significantly lower cost compared to traditional statistical feature-based models. This demonstrates that topological features not only enhance forecast accuracy but also provide a computationally efficient alternative when feature selection is omitted.

Figures 12, 13, and 14 illustrate comparisons between actual and predicted values over a four-hour forecasting horizon. In these visualizations, the right-hand side of each plot displays the test set performance, while the left-hand side shows model performance on the training set, allowing for a clear assessment of potential overfitting. The consistency between train and test results indicates that our model generalizes well without overfitting.

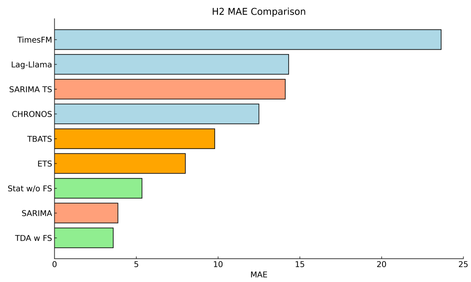
Lastly, Figure 15 provides a comparative performance overview of various forecasting models across the GH2, H2, and GM2 datasets. In these bar plots, blue denotes the newer time series models, while green represents our proposed model. The orange color indicates classical models such as TBATS and ETS, and reddish tones correspond to SARIMA-based models without exogenous variables. These colors are visually reflected in the plots to distinguish model families and help clarify relative performance. It is evident that our approach with topological features consistently achieves the lowest MAE and MAPE values, outperforming all other models across all datasets.



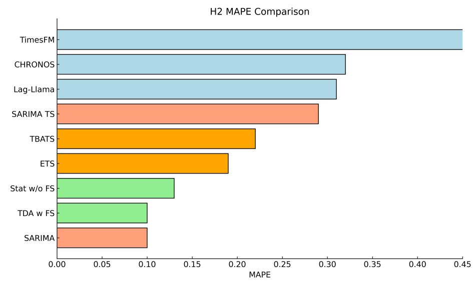
(a) MAE for GH2 Dataset



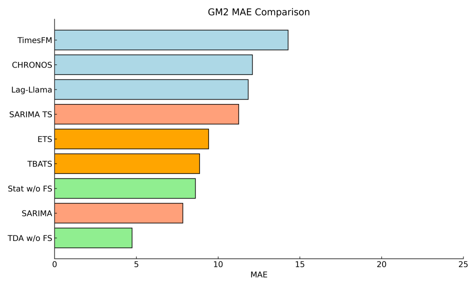
(b) MAPE for GH2 Dataset



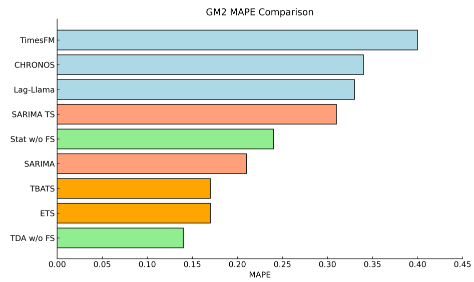
(c) MAE for H2 Dataset



(d) MAPE for H2 Dataset



(e) MAE for GM2 Dataset



(f) MAPE for GM2 Dataset

Fig. 15: MAE and MAPE comparisons for GH2, H2, and GM2 datasets.

6 Conclusion

In this study, we presented a novel method for forecasting short-term, highly volatile OEE data by incorporating topological features into time series analysis. The results demonstrate that the proposed approach can successfully predict OEE dynamics even in the face of significant volatility. A key factor in this success is the use of TDA to capture the underlying dynamics of the complex OEE time series, extracting features that reflect its intrinsic geometric and temporal structure. Additionally, decomposing the time series into its trend, seasonal, and residual components enabled the model to address each source of variation separately, further enhancing forecasting accuracy.

Furthermore, our approach benefited from a rigorous feature selection procedure that improved both model efficiency and accuracy. We applied RFE driven by feature p-values to filter out statistically insignificant predictors, followed by PSO guided by the BIC to identify an optimal subset of features. This two-step selection strategy eliminated redundant information and ensured that only the most informative features were used, thereby streamlining the model without sacrificing predictive performance.

In comparison to classical statistical models, which often struggle to forecast highly volatile series due to their rigid assumptions (e.g., linearity, stationarity), the proposed method demonstrates superior adaptability and precision. Traditional models such as ARIMA or exponential smoothing are limited in their ability to capture the complex, nonlinear patterns present in OEE data, whereas our approach augmented with topological insights and advanced feature selection can effectively model these complexities. This advantage makes the method particularly valuable for intelligent manufacturing management in an Industry 4.0 environment. Modern production systems generate vast amounts of data, and our data driven, feature-enhanced forecasting framework is well suited to leverage this data-rich context, providing decision makers with timely and reliable OEE predictions to support improved operational decisions.

Looking ahead, there are several promising directions to extend this research. One avenue is to apply the method to multivariate forecasting, where OEE is predicted alongside or informed by related operational metrics. By modeling multiple interrelated time series, the approach could capture cross dependencies and further improve predictive accuracy. Another avenue is to develop real time adaptive forecasting systems that continuously update model parameters as new data become available, thereby maintaining high accuracy under changing operational conditions. Pursuing these extensions would further increase the method's practicality and impact for smart manufacturing in dynamic Industry 4.0 settings.

7 Real World Application

In modern white good production facilities, OEE plays a critical role in enabling proactive operational management. This section presents a real-world application deployed in a dishwasher manufacturing plant, where an Application Programming Interface (API) has been implemented to monitor OEE values in real time across 14 distinct equipment units.

The application is built on a time series decomposition framework, enabling OEE values to be analyzed in terms of their trend, seasonal, and residual components. The decomposition captures multiple seasonalities associated with production operations, including 8-hour (shift-level), 24-hour (daily), and 168-hour (weekly) cycles. Figure 16 visualizes the user interface of the API, while Figures 17 and 18 illustrate the seasonal and residual components respectively.

The trend component reflects the long-term evolution of equipment performance, allowing managers to observe gradual efficiency improvements or degradation. Seasonal components, by contrast, highlight recurrent behavioral patterns, such as performance dips near shift changes or inefficiencies during night shifts. These patterns can be directly associated with operational characteristics such as human fatigue, handovers, or material availability. The residual component, which encapsulates stochastic

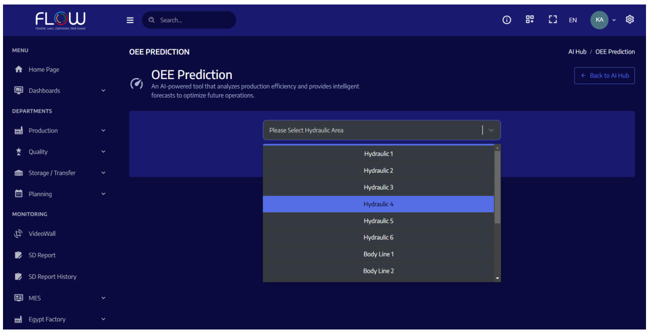


Fig. 16: API interface displaying equipment list that can be selected to forecast the OEE values for next 4 hours.

variations not explained by trend or seasonality, serves as an indicator of unanticipated performance losses. By continuously tracking residual behavior, the system provides early warnings of sudden anomalies, such as micro-downtimes or quality losses due to undetected machine faults. Furthermore, the forecasting capability of the application

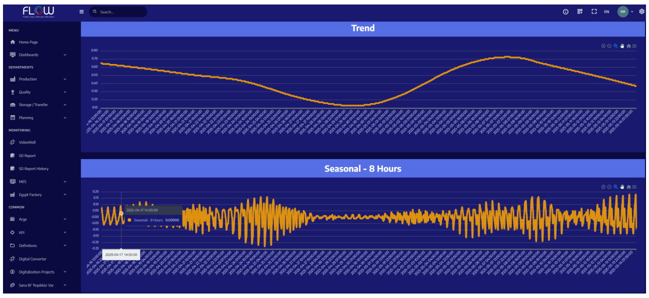


Fig. 17: Visualization of trend and 8-hour seasonality components of OEE time series.

extends to the upcoming shift, offering a probabilistic estimation of OEE. This foresight empowers production and maintenance teams to allocate resources proactively. When the system predicts a potential decline in OEE, maintenance interventions can be scheduled in advance, or production loads can be adjusted accordingly. Thus, the API supports a closed-loop, data-driven management system that continuously aligns production planning with real-time equipment behavior. This deployment illustrates the integration of time series forecasting techniques into an operational setting, demonstrating the effectiveness of decomposition-based OEE monitoring in achieving data-informed decision making and responsive production control.

Fig. 18: The forecast result on the API.

Declarations

- Funding: Not applicable
- Conflict of interest/Competing interests: The authors declare no competing interests.
- Ethics approval and consent to participate: Not applicable
- Consent for publication: Consent was obtained for publication of the equipment photos presented in Figures 3 and 4.
- Data availability: The datasets generated and analyzed during the current study are available in the following GitHub repository: https://github.com/korkutanapa/OEE_ARTICLE_STUDIES
- Materials availability: Not applicable
- Code availability: The code used for analysis and modeling in this study is available at: https://github.com/korkutanapa/OEE_ARTICLE_STUDIES
- Author contribution: All authors contributed equally to the conceptualization, methodology, analysis, and writing of this manuscript.

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